# Natural Language Processing
# Elective Course

## 1. Course Description

This course is designed to provide the fundamental principles and methodologies of Natural Language Processing (NLP). Students will learn to develop systems that can automatically interpret, analyze, and generate human language from textual data. The course covers essential techniques from text preprocessing and traditional machine learning models to advanced neural architectures, including word embeddings, recurrent networks, and the transformer models that power modern AI. A strong emphasis is placed on both theoretical understanding and practical implementation using state-of-the-art tools and libraries.

## 2. General Objectives

- To provide the students with key concepts and principles of Natural Language Processing (NLP) and its core challenges.
- To acquaint the students with the skills to develop and implement different NLP algorithms, from traditional models to modern neural approaches.
- To develop the skills in students to use popular NLP tools and frameworks (such as NLTK, spaCy, and Hugging Face) and apply NLP techniques to solve real-world problems. • To acquaint the students with the knowledge of advanced topics in neural language models, including Word Embeddings, Recurrent Neural Networks (RNNs), and the Transformer architecture.
- To provide the students with the knowledge to evaluate, interpret, and critically assess the performance and ethical implications of NLP models.

## 3. Methods of Instruction

Lecture, Discussion, Readings, Practical works and Project works.

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## 4. Contents in Detail

| Specific Objectives | Contents |
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| • Understand the concept of NLP in detail | **1. Introduction to NLP ( 4 hrs)**<br>**1.1.** What is NLP? Natural Language Understanding vs. Generation<br>**1.2.** Human language and intelligence (Turing Test) **1.3.** Applications of NLP in industry<br>**1.4.** Major challenges: ambiguity, context dependence, language diversity |

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| ● Implementation of text preprocessing methods and data collection method | **2. Text Preprocessing and Feature Engineering (8 hrs)**<br>**2.1.** Data collection and types of NLP data<br>**2.2.** Text cleaning: lowercasing, punctuation removal, whitespace handling<br>**2.3.** Tokenization: sentence and word tokenization **2.4.** Stop word removal, stemming, lemmatization |
| ● Implementation of text vectorization and basic encoding methods | **3. Traditional Text Representation (7 hrs) 3.1.** Label encoding and one-hot encoding<br>**3.2.** Bag of Words (BoW) model<br>**3.3.** TF-IDF: term frequency, inverse document frequency<br>**3.4.** BM25 for improved term weighting<br>**3.5.** Vector space models and cosine similarity |
| ● To understand the fundamental principles and limitations of classical vs. neural word representation methods. | **4.Word Embeddings and Neural Methods (6 hrs)**<br>4.1. Word2Vec: CBOW and Skip-gram architectures<br>4.2. GloVe: Global Vectors for Word Representation<br>4.3. FastText and subword information<br>4.4. Evaluating word embeddings |
| ● Apply the various techniques to evaluate and validate machine learning algorithms. | 5.**Text Classification and Language Models (5 hrs)** 5.1. Naive Bayes classifier with Laplace smoothing 5.2. n-gram language models: unigram, bigram, trigram<br>5.3. Neural language models<br>5.4. Evaluation metrics for classification |
| ● To implement probabilistic classifiers like Naive Bayes with smoothing techniques for foundational text classification tasks. | **6. Transformers and Modern Architectures** 6.1. Attention mechanism: self-attention, multi-head attention<br>6.2. Transformer architecture: encoder-decoder structure<br>6.3. Positional encoding and feedforward networks 6.4. BERT: Bidirectional Encoder Representations |

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| ● To develop the skills to effectively fine-tune large pre-trained language models for specific downstream NLP tasks and evaluate the ethical implications, societal biases, and privacy | **7. Advanced Applications and Ethics**<br>7.1 Masked Language Modeling (MLM)<br>7.2 Next Sentence Prediction (NSP)<br>7.3 Fine-tuning pretrained models<br>7.4 Ethical considerations: bias, fairness, privacy |

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| concerns inherent in large-scale language models. | |

**5. Practical Works**

Laboratory work of 30 hours per group of a maximum of 24 students must cover the following lab works:

| SN | Implementation Description |
|---|---|
| 1 | Setting up NLP development environment and applying basic nlp techniques |
| 2 | Building a complete text preprocessing pipeline |
| 3 | Document similarity and search system |
| 4 | Training custom word embeddings |
| 5 | Sentiment analysis system |
| 6 | Fine-tuning BERT for specific tasks |
| 7 | End-to-end NLP application development |

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Students must submit a project work that uses all the knowledge obtained from this course to solve any problem they choose. The marks for the practical evaluation must be based on the project work submitted by students.

**6. Evaluation System and Students' Responsibilities**

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**Evaluation System**

The internal evaluation of a student may consist of assignments, attendance, internal assessment, lab reports, project works etc. The internal evaluation scheme for this course is as follows:

| **Internal Evaluation** | **Weight** | **Marks** | **External Evaluation** | **Marks** |
|---|---|---|---|---|
| **Theory** | | 30 | | |
| Attendance & Class Participation | 10% | | | |
| Assignments | 20% | | | |
| Presentations/Quizzes | 10% | | Semester-End examination | 50 |
| Internal Assessment | 60% | | | |
| **Practical** | | 20 | | |
| Attendance & Class Participation | 10% | | | |
| Lab Report/Project Report | 20% | | | |
| Practical Exam/Project Work | 40% | | | |
| Viva | 30% | | | |
| **Total Internal** | | 50 | | |
| Full Marks: 50 + 50 = 100 | | | | |

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**Student Responsibilities**

Each student must secure at least 45% marks separately in internal assessment and practical evaluation with 80% attendance in the class in order to appear in the Semester End Examination. Failing to get such a score will be given NOT QUALIFIED (NQ) to appear for the Semester-End Examinations. Students are advised to attend all the classes, formal exam, test, etc. and complete all the assignments within the specified time period. Students are required to complete all the requirements defined for the completion of the course.

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## 7. Prescribed Books and References

### Text Books

**1.** “Speech and Language Processing” by Jurafsky and Martin.
**2.** Natural Language Processing with Transformers by Tunstall et al.

### References

**1.** Applied Natural Language Processing in the Enterprise" by Kulkarni and Shivananda

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